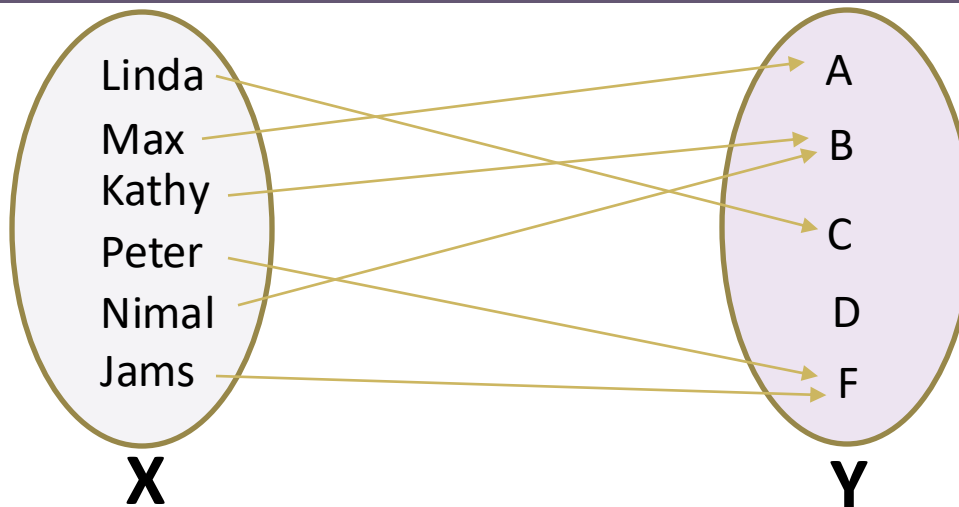


Lesson 6: Functions

- **Topics to be covered.**
 - **Defintion of a function**
 - **One-to-One Functions**
 - **Onto Functions**
 - **Inverse Functions**
 - **Composition of Functions**
 - **Some Important Functions**
 - **Partial Functions**

Functions

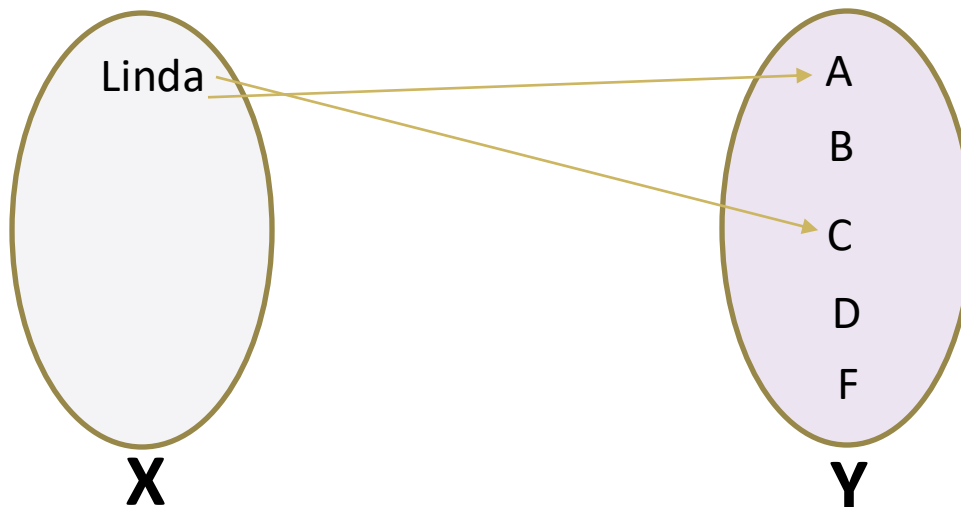


Definition: Let X and Y be nonempty sets. A function f from X to Y is an assignment of exactly one element of Y to each element of X . We write $f(a) = b$ if b is the unique element of Y assigned by the function f to the element a of X . If f is a function from X to Y , we write $f: X \rightarrow Y$.

Functions are sometimes also called mappings or transformations.

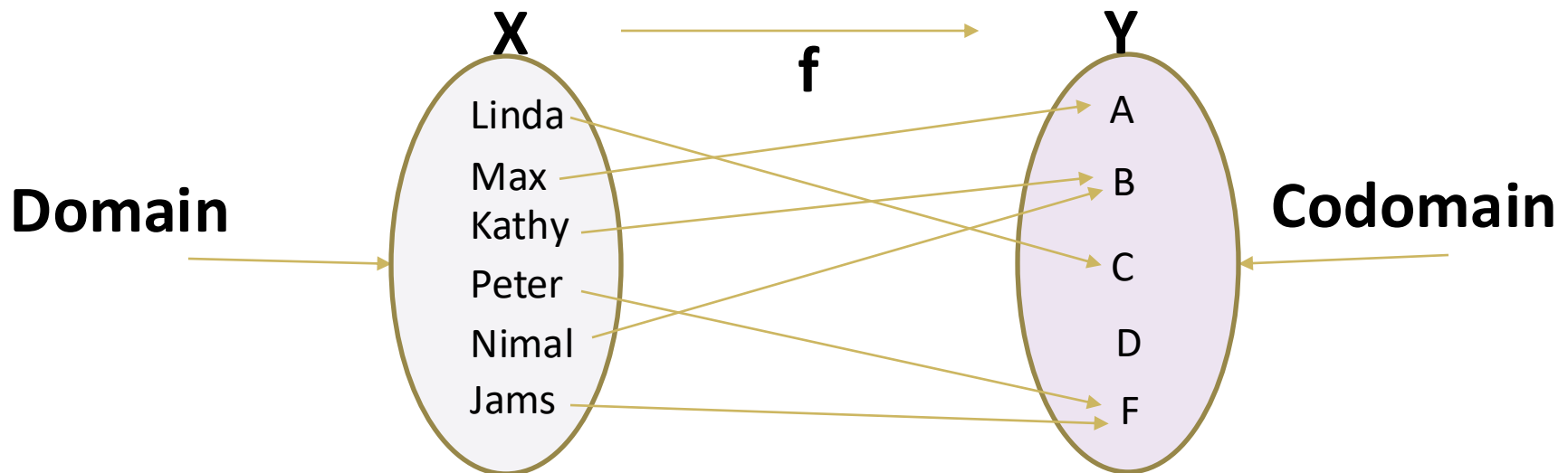
Requirement of functions

- An element of the set X cannot be mapped with two different elements in Y as shown in the following figure.



- Every element of the set x must be mapped with an element with Y subjected to the above condition.
- However there may be elements in y which are not assigned with elements of x from the function.

Terminology



If $f(a) = b$, we say that b is the **image** of a and a is a **pre-image** of b . The **range**, or image, of f denoted by $\text{Range}(f)$ is the set of all images of elements of X and is .

$$\text{Range}(f) = \{y \in Y \mid y = f(x) \text{ for some } x \in X\}.$$
$$\text{Range}(f) \subseteq \text{Codomain}(f)$$

Examples

Find the domain, codomain and range of the following functions:

1. Let f be the function that assigns the last two bits of a bit string of length 2 or greater to that string. For example, $f(11010) = 10$.
 - The domain of f is the set of all bit strings of length 2 or greater.
 - The codomain and range are the set $\{00,01,10,11\}$.
2. Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ assign the square of an integer to this integer. Then, $f(x) = x^2$, for all $x \in \mathbb{Z}$.
 - The domain of f is the set of all integers ($\mathbb{Z}$)
 - The codomain of f is the set of all integers ($\mathbb{Z}$)
 - The range of f is the set $\{n^2 \mid n \in \mathbb{Z}\} = \{0, 1, 4, 9, 16, \dots\}$.

One to one function

Some functions never assign the same co-domain value to two different domain elements. These functions are said to be one-to-one.

Definition: A function f is said to be one-to-one, or an injection, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f .

Remark: We can express that f is one-to-one using quantifiers as follows:

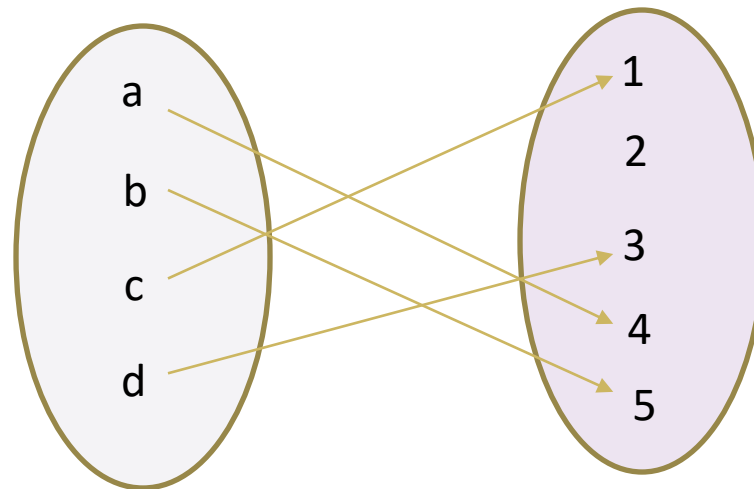
$$f \text{ is one-to-one} \Leftrightarrow \forall a, \forall b (f(a) = f(b) \rightarrow a = b)$$

$$\Leftrightarrow \forall a, \forall b (a \neq b \rightarrow f(a) \neq f(b)),$$

where the universe of discourse is the domain of the function.

Example

Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ is one-to-one.



The function f is **one to one** because f assigns different values to the elements of its domain.

Examples

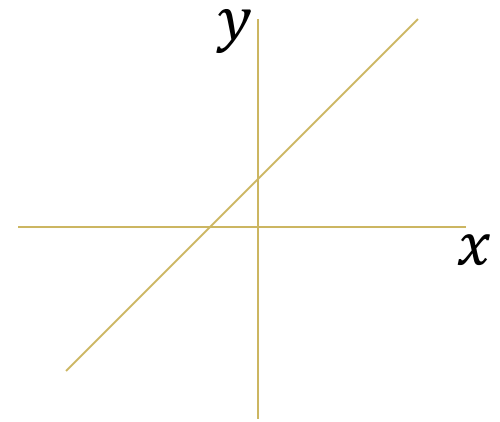
- Determine whether the function $f(x) = x + 1$ from the set of real numbers to itself is one-to-one.

Suppose that $f(x_1) = f(x_2)$ for some $x_1, x_2 \in \mathbb{R}$.

$$\Rightarrow x_1 + 1 = x_2 + 1$$

$$\Rightarrow x_1 = x_2.$$

Hence, f is one to one.



- Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one-to-one.

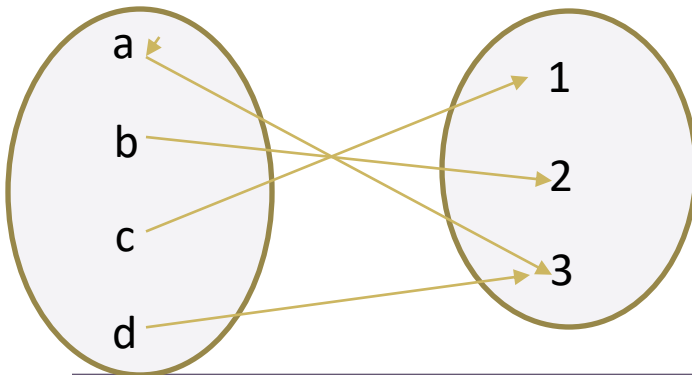
Since $f(-1) = f(1) = 1$, and $-1 \neq 1$, it follows that f is not one to one.

Onto Functions

Definition: A function f from A to B is called **onto**, or a **surjection**, if and only if for every element $b \in B$ there is an element $a \in A$ such that $f(a) = b$. That is;

$$\begin{aligned} f: A \rightarrow B \text{ is onto} &\Leftrightarrow \forall y \in B \exists x \in A \text{ such that } f(x) = y. \\ &\Leftrightarrow f(A) = \text{Range}(f) = B = \text{Co-domain}(f) \end{aligned}$$

Example: Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3, f(b) = 1, f(c) = 2, f(d) = 3$. Is f an onto function?



Because all three elements of the codomain are images of elements in the domain, f is an onto function.

Examples

- Is the function $f(x) = 2x + 1$ from the set of real numbers to the set of real numbers onto?

Solution: Let $y \in \mathbb{R}$ be arbitrary.

Then $x = \frac{y-1}{2} \in \mathbb{R}$ and $f(x) = y$

$$f(x) = f\left(\frac{y-1}{2}\right) = y - 1 + 1 = y.$$

Hence, $\forall y \in \mathbb{R} \exists x \in \mathbb{R}$ such that $f(x) = y$.

Therefore f is onto.

$$\begin{aligned} f(x) &= y \\ \Rightarrow 2x + 1 &= y \\ \Rightarrow x &= \frac{y-1}{2} \end{aligned}$$

- Is the function $f(x) = x^2$ from the set of integers to the set of integers onto?

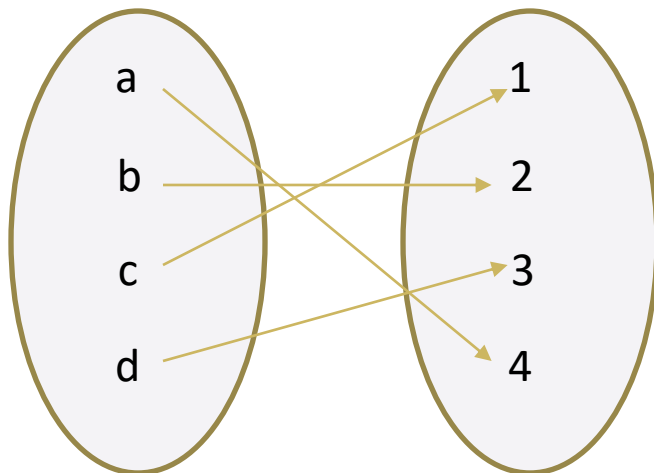
The function f is not onto because there is no integer x with $x^2 = -1$.

One to one correspondence or Bijection

Definition: A function $f: A \rightarrow B$ is a bijection, if and only if it is both one to one and onto.

Example:

Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a) = 4, f(b) = 2, f(c) = 1, f(d) = 3$. Is f a bijection?



The function f is one to one and onto.

Hence, f is a bijection.

Example

Consider the function $f: \mathbb{R} - \{3\} \rightarrow \mathbb{R} - \{1\}$ defined by $f(x) = \left(\frac{x-2}{x-3}\right), \forall x \in \mathbb{R} - \{3\}$. Is f a bijection? Justify your answer.

Suppose that $f(x) = f(y)$ for some $x, y \in \mathbb{R} - \{3\}$.

$$\Rightarrow \frac{x-2}{x-3} = \frac{y-2}{y-3}$$

$$\Rightarrow (x-2)(y-3) = (x-3)(y-2)$$

$$\Rightarrow xy - 3x - 2y + 6 = xy - 2x - 3y + 6$$

$$\Rightarrow x = y.$$

$\therefore f$ is one to one.

Let $y \in \mathbb{R} - \{1\}$ and $f(x) = y$

Since $y \in \mathbb{R} - \{1\}$ $y \neq 1$.

Since $f(x) = y$

$$\Rightarrow \frac{x-2}{x-3} = y$$

$$\Rightarrow (x-2) = (x-3)y = xy - 3y$$

$$\Rightarrow x - xy = 2 - 3y$$

$$\Rightarrow x(1-y) = 2 - 3y$$

$$\Rightarrow x = \frac{(2-3y)}{(1-y)} \in \mathbb{R} \quad (\because y \neq 1).$$

$\therefore \forall y \in \mathbb{R} - \{1\}, \exists x \in \mathbb{R} - \{3\}$ s.t $f(x) = y$.

Hence, f is onto

Since f is one to one and onto, f is a bijection.

Exercise

Give an example of a function from $\mathbb{Z}^+$ to $\mathbb{Z}^+$ that is

- one-to-one but not onto:

Define $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ by $f(n) = n + 1, \forall n \in \mathbb{Z}^+$.

Then f is one to one but not onto.

There is no number in $\mathbb{Z}^+$ (*domain*) that maps to 1 in the codomain

- onto but not one-to-one:

Define $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ by $f(n) = \begin{cases} 1 & \text{if } n = 1 \\ n - 1 & \text{if } n > 1 \end{cases}$.

Then f is not one to one, but f is onto.

$$f(2) = 2 - 1 = 1 = f(1)$$

Exercise

Give an example of a function from $\mathbb{Z}^+$ to $\mathbb{Z}^+$ that is

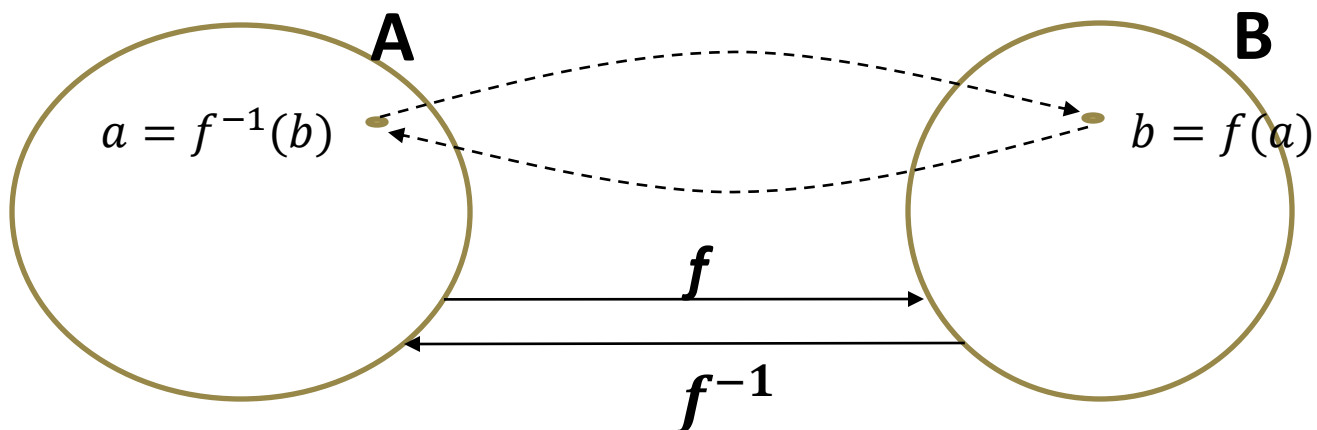
- both onto and one-to-one (but different from the identity function)

Define $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ by $f(n) = \begin{cases} n + 1 & \text{if } n \text{ is odd} \\ n - 1 & \text{if } n \text{ is even} \end{cases}$.

Inverse Functions

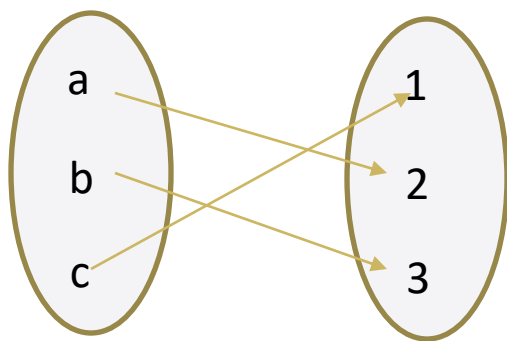
An interesting property of bijections is that they have an **inverse function**.

Definition: Let f be a bijection from the set A to the set B . The inverse function of f is the function that assigns to an element b belonging to B a unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$.

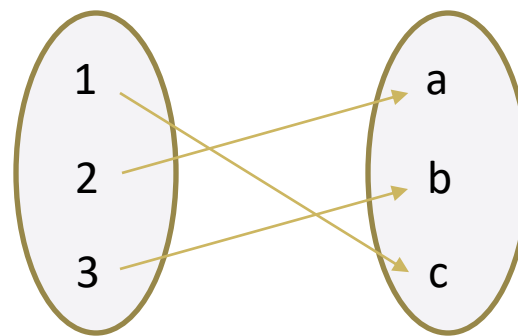


Example

Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?



$f: A \rightarrow B$



$f^{-1}: B \rightarrow A$

Examples

Let f be the function from $\mathbb{R}$ to $\mathbb{R}$ with $f(x) = x^2$. Is f invertible?

Because $f(-1) = f(1) = 1$, f is not one-to-one. If f has an inverse function it would have to assign two elements 1 and -1 to 1. Hence, f is not invertible.

Examples

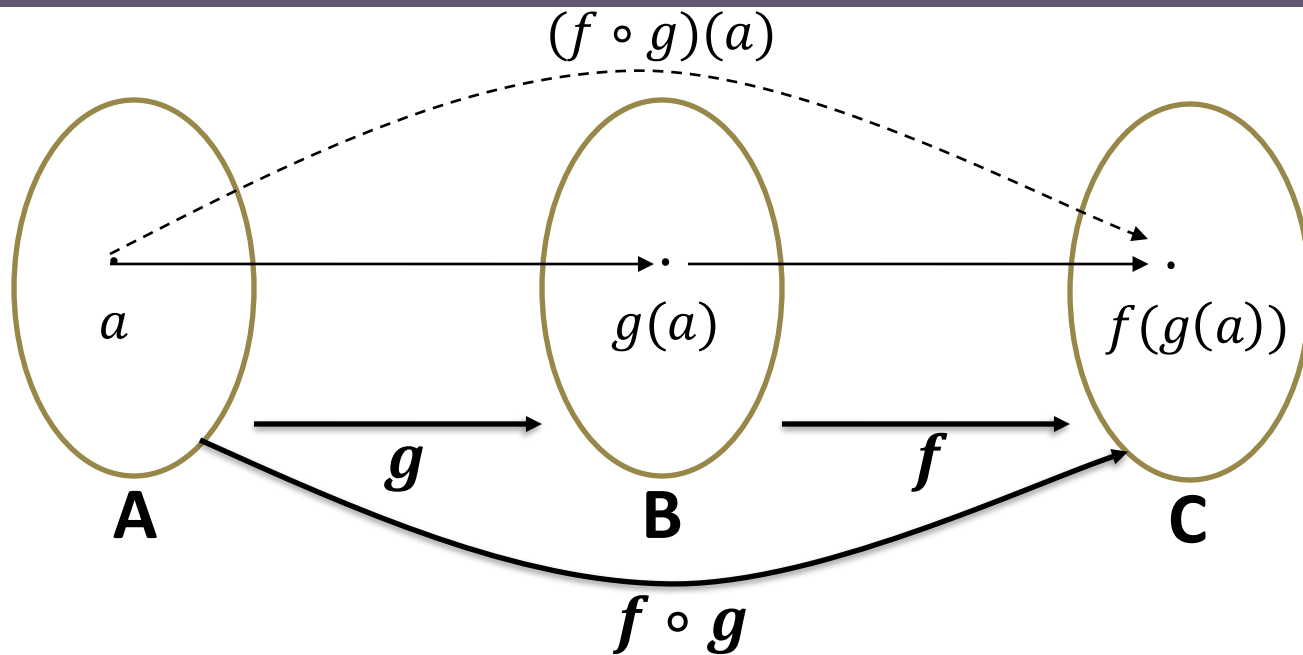
Let $f: \mathbb{R}^* \rightarrow \mathbb{R}^*$ be the function defined by $f(x) = x^2$, where $\mathbb{R}^* = \{x \in \mathbb{R} \mid x \geq 0\}$. Is f invertible?

Let $f(x) = f(y)$, for $x, y \in \mathbb{R}^*$
 $\Rightarrow x^2 = y^2$
 $\Rightarrow (x - y)(x + y) = 0$
 $\Rightarrow x = y, \text{ or } x = -y$
 $\Rightarrow x = y \ (\because x, y \geq 0)$
Hence, f is one to one.

Let $y \in \mathbb{R}^*$. Then $y \geq 0$.
 $\Rightarrow \exists x \in \mathbb{R}^*$ such that $x = \sqrt{y}$
 $\Rightarrow f(x) = x^2 = y$.
That is; $\forall y \in \mathbb{R}^*, \exists x \in \mathbb{R}^*$ s.t $f(x) = y$.
Hence, f is onto.

Since, f is one to one and onto f is invertible and
 $f^{-1}(y) = \sqrt{y}, \forall y \in \mathbb{R}^*$.

Compositions of Functions



Definition: The **composition** of two functions $g: A \rightarrow B$ and $f: B \rightarrow C$, denoted by $f \circ g$, is defined by

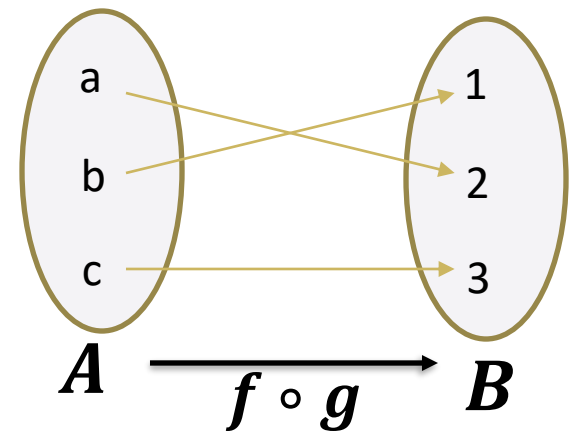
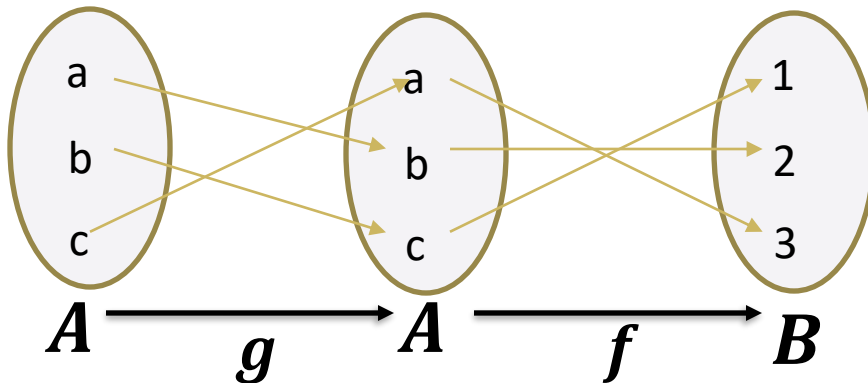
$$(f \circ g)(a) = f(g(a)) \text{ for all } a \in A.$$

Note that the composition $f \circ g$ cannot be defined unless the range of g is a subset of the domain of f .

Example

Let g be the function from the set $\{a,b,c\}$ to itself such that $g(a) = b$, $g(b) = a$, and $g(c) = c$. Let f be the function from the set $\{a,b,c\}$ to the set $\{1,2,3\}$ such that $f(a) = 1$, $f(b) = 2$, and $f(c) = 3$.

1. What is the composition of f and g , and what is the composition of g and f ?



Note that $g \circ f$ is not defined, because the range of $f = \{1, 2, 3\}$ is not a subset of the domain of $g = \{a, b, c\}$.

Example

Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?

Let $x \in \mathbb{Z}$ be arbitrary.

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(3x + 2) \\ &= 2(3x + 2) + 3 \\ &= 6x + 7.\end{aligned}$$

Let $x \in \mathbb{Z}$ be arbitrary.

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(2x + 3) \\ &= 3(2x + 3) + 2 \\ &= 6x + 11.\end{aligned}$$

Note that even though $f \circ g$ and $g \circ f$ are defined for the functions f and g , $f \circ g$ and $g \circ f$ are not equal.

More Examples

Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be both one to one functions. Show that $g \circ f: A \rightarrow C$ is one to one.

Solution: Assume that $(g \circ f)(a) = (g \circ f)(b)$ for $a, b \in A$.

$$\Rightarrow g(f(a)) = g(f(b)) \quad (\because \text{Def. of } g \circ f)$$

$$\Rightarrow f(a) = f(b) \quad (\because g \text{ is } 1-1)$$

$$\Rightarrow a = b \quad (\because f \text{ is } 1-1)$$

Hence, $\forall a, b \in A \quad [(g \circ f)(a) = (g \circ f)(b) \rightarrow a = b]$

Therefore $g \circ f$ is one-to-one.

Exercise: Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be both onto functions. Show that $g \circ f: A \rightarrow C$ is onto.

Identity Function

Definition: Let A be a non-empty set. The **identity function** on A is the function $i_A: A \rightarrow A$, where $i_A(x) = x$ for all $x \in A$. In other words, the identity function i_A is the function that assigns each element to itself.

The function i_A is one-to-one and onto, so it is a bijection.

When the composition of a function $f: A \rightarrow B$ and its inverse, if exists $f^{-1}: B \rightarrow A$ is formed, in either order, an identity function is obtained.

$$\begin{aligned}(f^{-1} \circ f)(a) &= f^{-1}(f(a)) = a \text{ for all } a \in A, \text{ and} \\ &= i_A(a) \\ (f \circ f^{-1})(b) &= f(f^{-1}(b)) = b \text{ for all } b \in B. \\ &= i_B(b)\end{aligned}$$

Hence $f^{-1} \circ f = i_A$, $f \circ f^{-1} = i_B$ and $(f^{-1})^{-1} = f$.

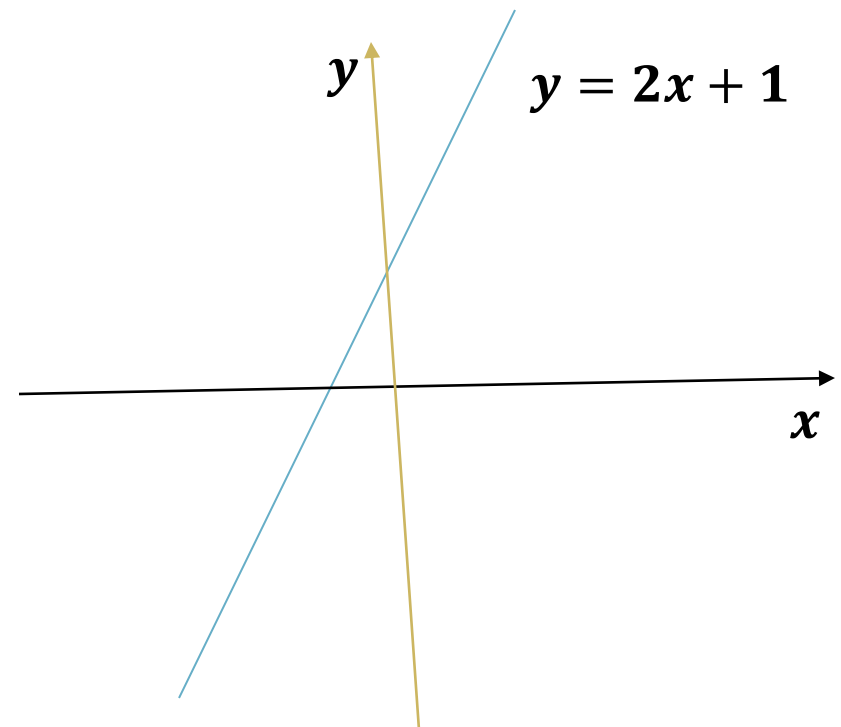
The Graphs of Functions

Definition: Let f be a function from the set A to the set B . The graph of the function f is denoted by $\mathcal{G}(f)$, and is the set of ordered pairs

$$\mathcal{G}(f) = \{(a, b) \mid a \in A \text{ and } f(a) = b\}.$$

Example: Display the graph of the function $f(x) = 2x + 1$ from the set of real numbers to the set of real numbers.

$$\begin{aligned}\mathcal{G}(f) &= \{(x, y) \mid x \in \mathbb{R}, y = 2x + 1\} \\ &= \{(x, 2x + 1) \mid x \in \mathbb{R}\}.\end{aligned}$$



The Image and Inverse Image of a Subset of the Domain & the Codomain Respectively

Let $f: A \rightarrow B$ be a function and let $S \subseteq A$ and $T \subseteq B$.

- The image of S under f is the subset of B that consists of the images of the elements of S , and is denoted by $f(S)$.

$$f(S) = \{ t \mid \exists s \in S (t = f(s)) \} = \{ f(s) \mid s \in S \}$$

Example -2

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = x^2, \forall x \in \mathbb{R}$.
Find

1. $f(\{-2, -1, 0, 1, 2\}) = \{0, 1, 4\}$
2. $f(\mathbb{Z}) = \{f(n) \mid n \in \mathbb{Z}\} = \{n^2 \mid n \in \mathbb{Z}\} = \{n^2 \mid n \in \mathbb{N}\}.$
3. $f(\mathbb{R}) = \{f(x) \mid x \in \mathbb{R}\} = \{x^2 \mid x \in \mathbb{R}\}$
4. $f^{-1}(\{1\}) = \{-1, 1\}$
5. $f^{-1}(\{x \mid 0 < x < 1\}) = \{x \mid -1 < x < 0 \text{ or } 0 < x < 1\}.$

Exercise -1

Let f be a function from the set A to the set B . Let S and T be subsets of A . Show that

1. $f(S \cup T) = f(S) \cup f(T)$
2. $f(S \cap T) \subseteq f(S) \cap f(T)$.

Solution for 1:

Let $y \in f(S \cup T)$.

$\Rightarrow \exists x \in (S \cup T)$ s.t $f(x) = y$.

$\Rightarrow \exists x \in S$ s.t $f(x) = y$ or

$\exists x \in T$ s.t $f(x) = y$

$\Rightarrow y \in f(S)$ or $y \in f(T)$

$\Rightarrow y \in f(S) \cup f(T)$

$\therefore f(S \cup T) \subseteq f(S) \cup f(T)$.

Let $y \in f(S) \cup f(T)$.

$\Rightarrow y \in f(S)$ or $y \in f(T)$

$\Rightarrow \exists x \in S$ s.t $f(x) = y$ or

$\exists x \in T$ s.t $f(x) = y$

$\Rightarrow \exists x \in (S \cup T)$ s.t $f(x) = y$

$\Rightarrow y \in f(S \cup T)$

$\therefore f(S) \cup f(T) \subseteq f(S \cup T)$.

Solution for 2:

Let $y \in f(S \cap T)$.

$\Rightarrow \exists x \in (S \cap T)$ s.t $f(x) = y$.

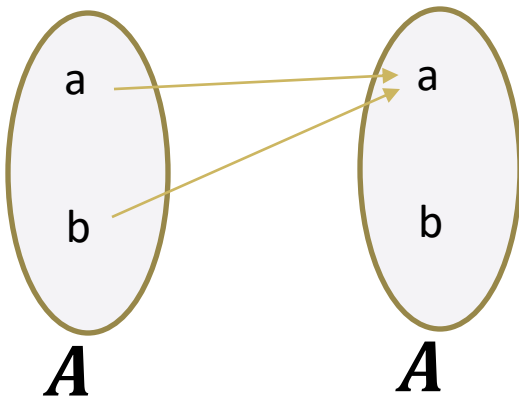
$\Rightarrow \exists x \in S$ s.t $f(x) = y$ and $\exists x \in T$ s.t $f(x) = y$

$\Rightarrow y \in f(S)$ and $y \in f(T)$

$\Rightarrow y \in f(S) \cap f(T)$

$\therefore f(S \cap T) \subseteq f(S) \cap f(T)$.

Let f be the function defined by the following diagram:



Let $S = \{a\}$ and $T = \{b\}$. Then

$S \cap T = \varnothing$, $f(S) = \{a\}$, and $f(T) = \{a\}$.

But $f(S \cap T) = \varnothing$, and $f(S) \cap f(T) = \{a\}$.

Exercise -2

Let f be a **one-to-one** function from the set A to the set B . Let S and T be subsets of A . Show that $f(S \cap T) = f(S) \cap f(T)$.

Solution: We know from exercise -1 that $f(S \cap T) \subseteq f(S) \cap f(T)$ when f is a function from A to B .

Now we shall show that $f(S) \cap f(T) \subseteq f(S \cap T)$ when f is one-to-one.

Let $y \in f(S) \cap f(T)$

$\Rightarrow y \in f(S)$ and $y \in f(T)$

$\Rightarrow \exists s \in S$ s.t $y = f(s)$ and $\exists t \in T$ s.t $y = f(t)$

$\Rightarrow \exists s \in S$ and $\exists t \in T$ s.t $y = f(s) = f(t)$

$\Rightarrow s = t$ and $s \in (S \cap T)$ and $y = f(s)$ ($\because f$ is one – to – one)

$\Rightarrow y \in f(S \cap T)$

$\therefore f(S) \cap f(T) \subseteq f(S \cap T)$

Example

Let f be a function from A to B . Let S and T be subsets of B . Show that

- a) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$
- b) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$
- c) $f^{-1}(B - S) = A - f^{-1}(S)$.

a)

Let $x \in f^{-1}(S \cup T)$

$$\Leftrightarrow f(x) \in (S \cup T)$$

$$\Leftrightarrow f(x) \in S \text{ or } f(x) \in T$$

$$\Leftrightarrow x \in f^{-1}(S) \text{ or } x \in f^{-1}(T)$$

$$\Leftrightarrow x \in (f^{-1}(S) \cup f^{-1}(T))$$

$$\therefore f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$$

c)

Let $x \in f^{-1}(B - S)$

$$\Leftrightarrow f(x) \in (B - S)$$

$$\Leftrightarrow f(x) \in B \text{ and } f(x) \notin S$$

$$\Leftrightarrow x \in f^{-1}(B) \text{ and } x \notin f^{-1}(S)$$

$$\Leftrightarrow x \in A \text{ and } x \notin f^{-1}(S)$$

$$\Leftrightarrow x \in A - f^{-1}(S).$$

$$\therefore f^{-1}(B - S) = A - f^{-1}(S).$$

Special Functions: Floor Function

The **floor** function $\lfloor \cdot \rfloor : \mathbb{R} \rightarrow \mathbb{Z}$ assigns to the real number x the largest integer that is less than or equal to x . The value of a floor function at x is denoted by $\lfloor x \rfloor$.

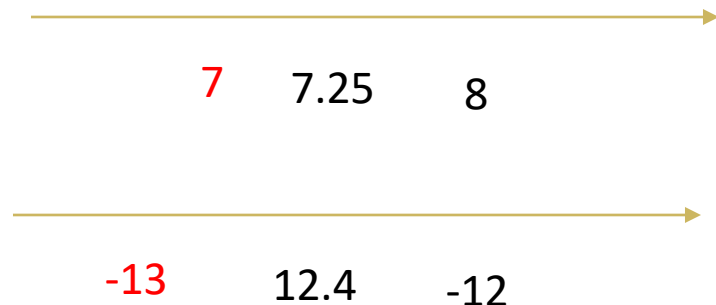
The floor function is often also called the greatest integer function.

$$\lfloor 0.5 \rfloor = 0$$

$$\lfloor 7.25 \rfloor = 7$$

$$\lfloor -12.4 \rfloor = -13$$

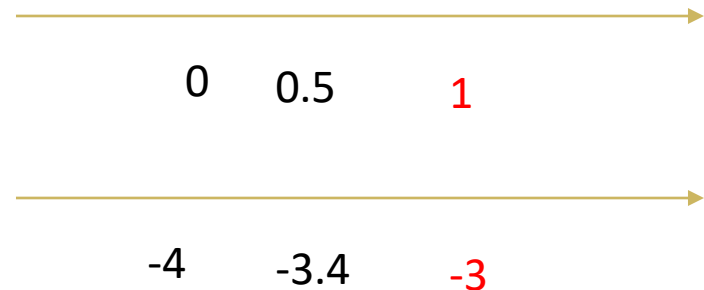
$$\lfloor -10 \rfloor = -10.$$



Special Functions: Ceiling Function

The **Ceiling** function $\lceil \cdot \rceil: \mathbb{R} \rightarrow \mathbb{Z}$ assigns to the real number x the smallest integer that is greater than or equal to x . The value of a ceiling function at x is denoted by $\lceil x \rceil$.

- $\lceil 0.5 \rceil = 1$
- $\lceil 7.5 \rceil = 8$
- $\lceil -3.4 \rceil = -3$.



Partial Functions

Definition: A partial function f from a set A to a set B is an assignment to each element a in a subset of A , called the domain of definition of f , of a unique element b in B . The sets A and B are called the domain and codomain of f , respectively. We say that f is undefined for elements in A that are not in the domain of definition of f . When the domain of definition of f equals A , we say that f is a total function.

Example: The function $f : Z \rightarrow R$ where $f(n) = \sqrt{n}$ is a partial function from Z to R where the domain of definition is the set of nonnegative integers. Note that f is undefined for negative integers.

$$f(n) = \sqrt{n} \text{ if } n \geq 0$$